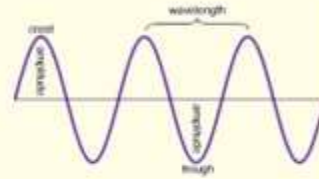
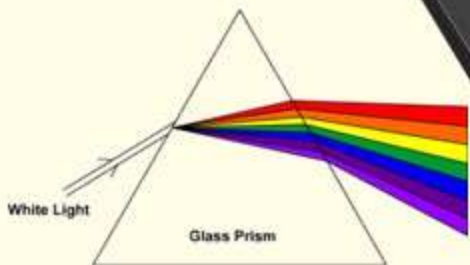


PHYSICS

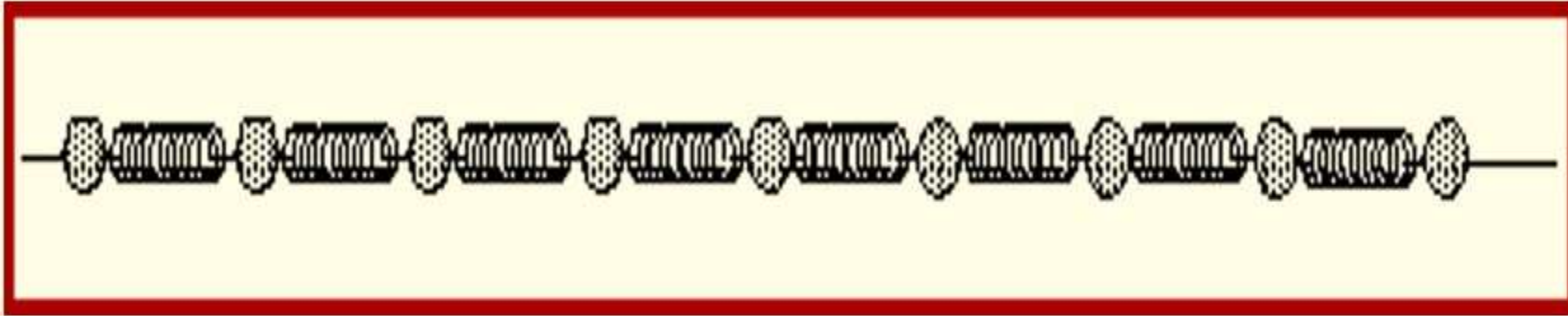
WAVES



MODULE - 12

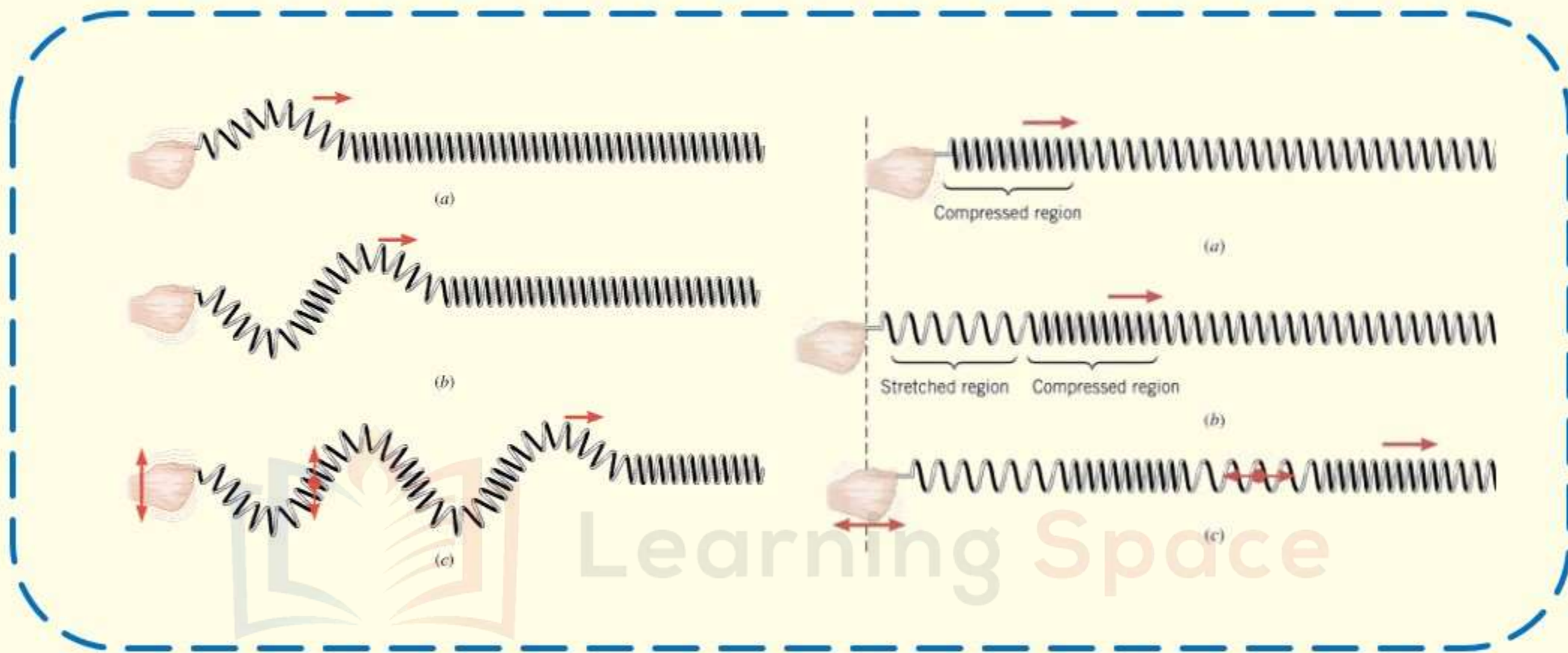


WAVE- MEANING IN PHYSICS



- *The repeating and periodic disturbance(created by a vibration) that moves through a medium from one location to another is referred to as a wave.*
- *A medium is a substance or material that carries the wave.*

WHAT TYPES OF WAVES CAN WE SEE?

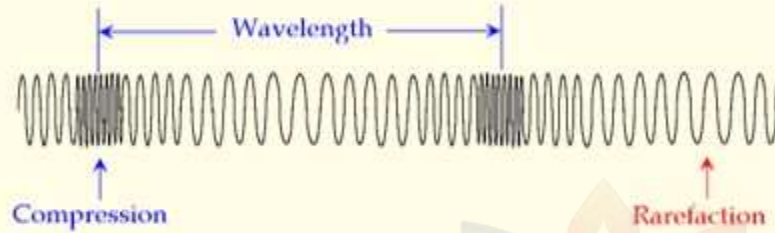


Do all waves travel the same way?

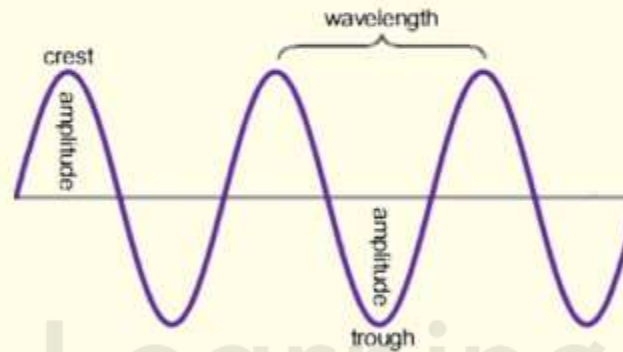


WAVES

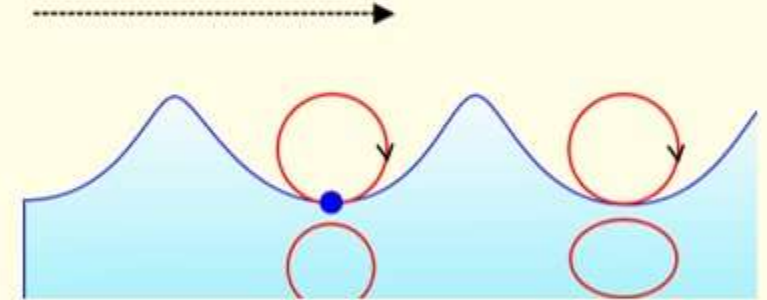
LONGITUDINAL WAVE



TRANSVERSE WAVE



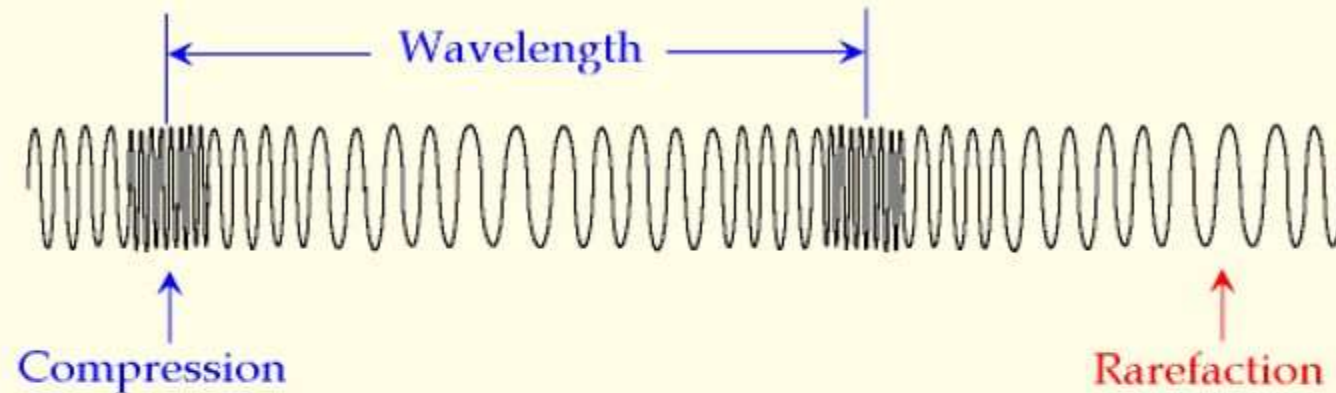
SURFACE WAVE



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LONGITUDINAL WAVE - ANATOMY



COMPRESSION

A region where the coils are pressed together in a small amount of space is known as a compression.

A compression is a point on a medium through which a longitudinal wave is traveling that has the maximum density.

RAREFACTION

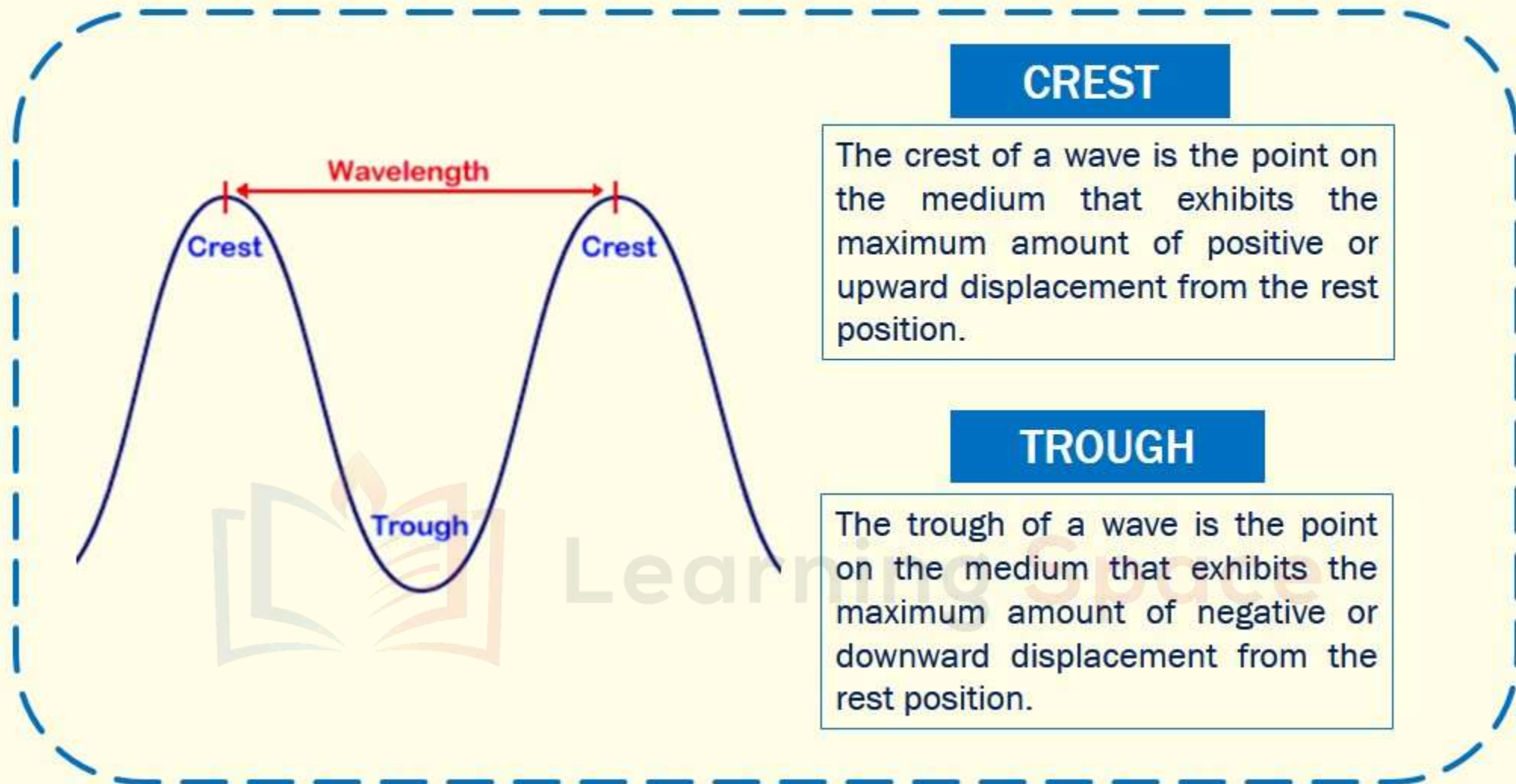
A region where the coils are spread apart, thus maximizing the distance between coils, is known as a rarefaction.

A rarefaction is a point on a medium through which a longitudinal wave is traveling that has the minimum density.

a wavelength measurement is made by measuring the distance from a compression to the next compression or from a rarefaction to the next rarefaction



TRANSVERSE WAVE - ANATOMY



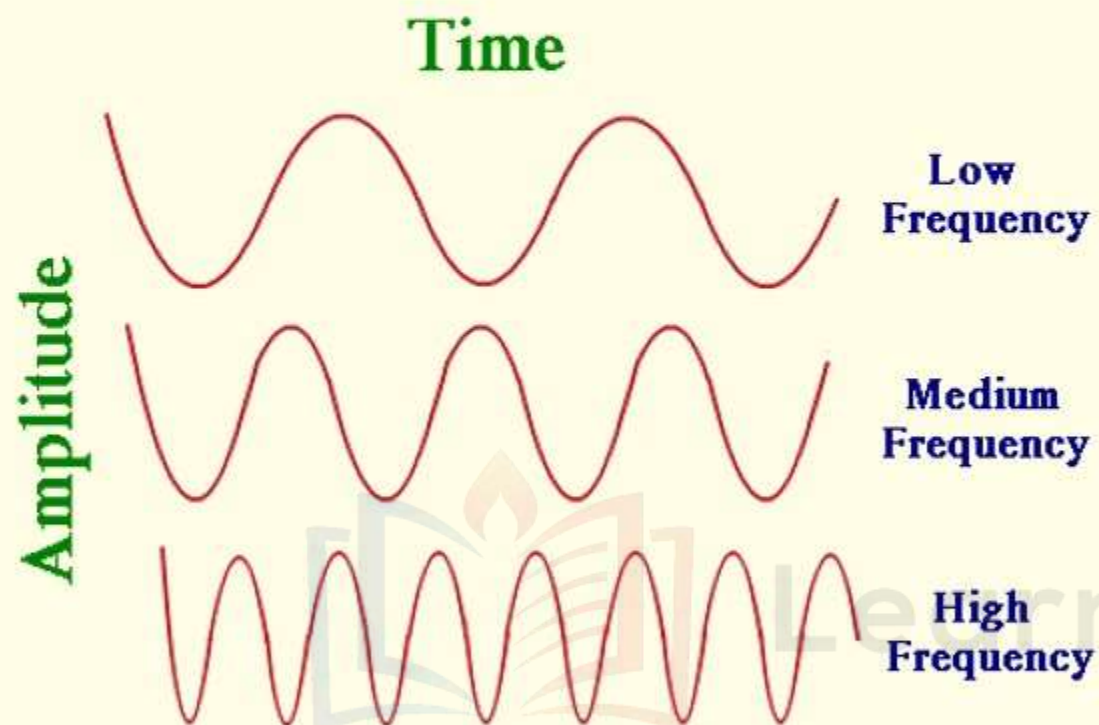
CREST

The crest of a wave is the point on the medium that exhibits the maximum amount of positive or upward displacement from the rest position.

TROUGH

The trough of a wave is the point on the medium that exhibits the maximum amount of negative or downward displacement from the rest position.

TRANSVERSE WAVE - ANATOMY



AMPLITUDE

Amplitude of a wave is the maximum amount of displacement of a wave /particle from rest to crest or from rest to trough.

FREQUENCY

The number of times the wavelength occurs in one second. SI Unit - Hz

TIME PERIOD

The time taken for a wave to complete a complete vibrational cycle, from one crest to another.

It is reciprocal to the frequency. SI Unit - seconds.



CONTRASTING FEATURE

LONGITUDINAL WAVE

- The particles of medium vibrate in the same direction.
- They are possible in all kinds of media.
- They consist of regions of compression and rarefaction.
- They cannot be polarised.
- Sound waves in air is an example of longitudinal waves.

TRANSVERSE WAVE

- The particles move at right angles to the direction of wave propagation.
- They are possible only in solids.
- They consist of crests and troughs.
- They can be polarised.
- Vibrations in a string is an example of transverse waves.





QUESTION 1

A transverse wave is transporting energy from east to west. The particles of the medium will move_____.

- a) east to west only
- b) both eastward and westward
- c) north to south only
- d) both northward and southward.

ANSWER : The particles would be moving back and forth in a direction perpendicular to energy transport. The waves are moving westward, so the particles move northward and southward.





QUESTION 2

If you strike a horizontal rod vertically from above, what can be said about the waves created in the rod?

- a. The particles vibrate horizontally along the direction of the rod.
- b. The particles vibrate vertically, perpendicular to the direction of the rod
- c. The particles vibrate in circles, perpendicular to the direction of the rod.
- d. The particles travel along the rod from the point of impact to its end

ANSWER : The particles vibrate in the direction of the source which creates the initial disturbance. Since the hammer was moving vertically, the particles will also vibrate vertically





QUESTION 3

As the frequency of an wave is doubled, it's time period will become?

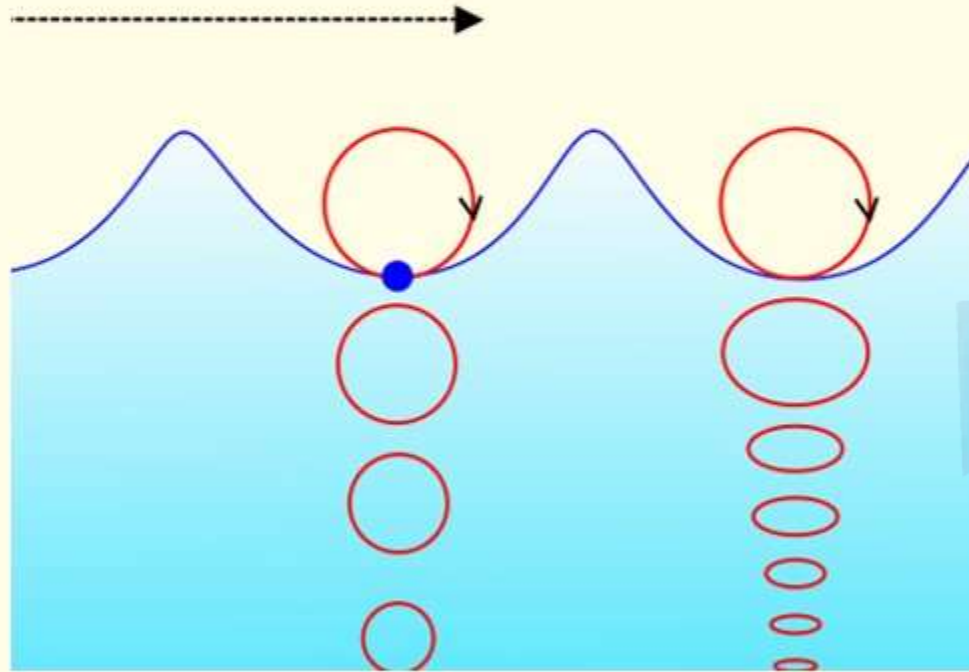
- a. doubled
- b. halved
- c. No change
- d. Quadripled

ANSWER :Because the frequency is reciprocal to its time period, when the frequency doubles, the time period reduces to half its original value.





SURFACE WAVE



- While waves that travel within the depths of the ocean are longitudinal waves, the waves that travel along the surface of the oceans are referred to as surface waves.
- A surface wave is a wave in which particles of the medium undergo a circular motion.
- Surface waves are neither longitudinal nor transverse.
- In a surface wave, it is only the particles at the surface of the medium that undergo the circular motion.
- The motion of particles tends to decrease as one proceeds further from the surface.



WAVES IN ORDER OF ABILITY TO TRANSMIT ENERGY

ELECTROMAGNETIC WAVE

- Do not require material medium for their propagation.
- Formed due to varying electric and magnetic fields.
- Speed of electromagnetic waves is high 3×10^8 m/s
- Transverse in nature.
- All light waves are examples of electromagnetic waves.

MECHANICAL WAVE

- Require material medium for their propagation.
- Formed due to vibrations of the particles of the medium.
- Speed of the sound waves is low, 332m/s.
- Can be transverse or longitudinal in nature.
- A sound wave is an example of a mechanical wave.





QUESTION 4

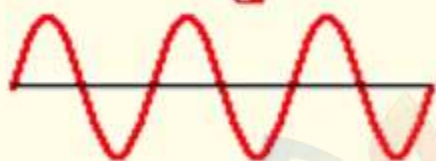
Which of the following is not a characteristic of mechanical waves?

- a. They consist of disturbances or oscillations of a medium.
- b. They transport energy without a medium
- c. They travel in a direction that is at right angles to the direction of the particles of the medium.
- d. They are created by a vibrating source

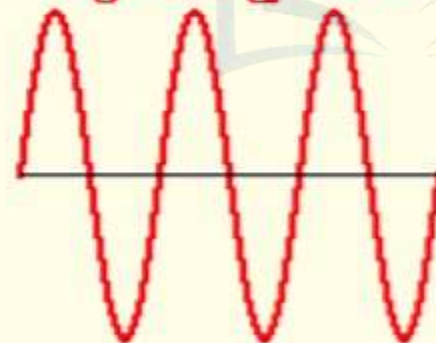


ENERGY OF A WAVE

Low-Energy Wave



High-Energy Wave



- The amount of energy carried by a wave is related to the amplitude of the wave. A high energy wave is characterized by a high amplitude; a low energy wave is characterized by a low amplitude.
- The energy transported by a wave is directly proportional to the square of the amplitude of the wave.

$$E \propto A^2$$

If the amplitude of the wave doubles, the energy quadruples





QUESTION 5

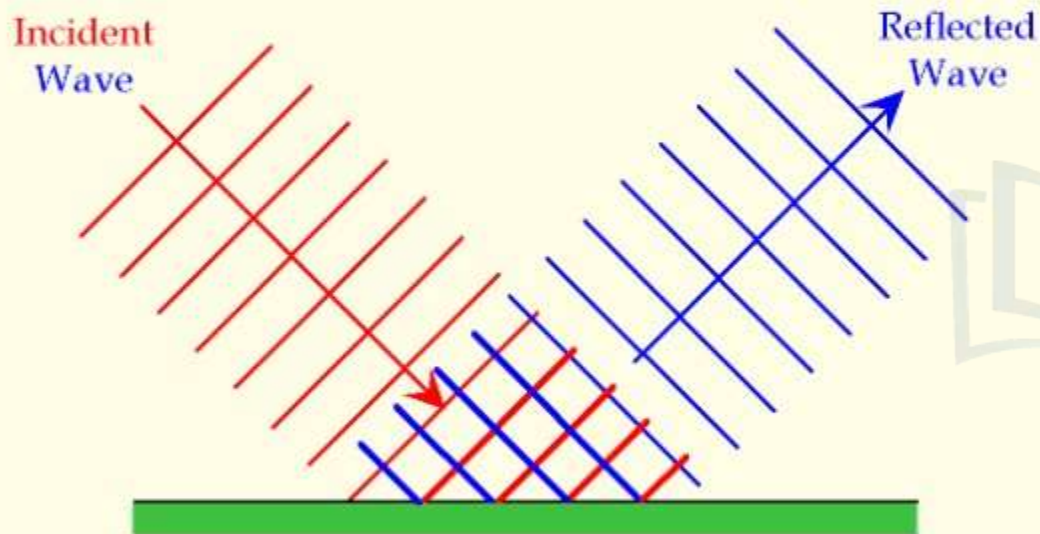
An ocean wave has an amplitude of 2.5 m. Weather conditions suddenly change such that the wave has an amplitude of 5.0 m. The amount of energy transported by the wave became?

- a. doubled.
- b. quadrupled
- c. halved
- d. One fourth of the original value

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REFLECTION IN A WAVE

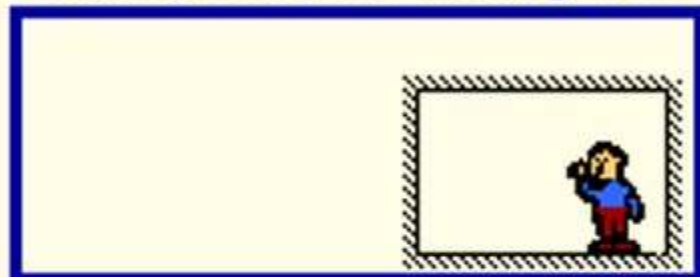


- A wave is a disturbance that moves along a medium from one end to the other.
- When a wave undergoes reflection, it remains within the medium and merely reverses its direction of travel.
- By reflecting back to the original location, the wave has traveled a distance that is equal to twice the length.
- Here, by the knowledge of the speed of the wave and the time taken to reach back, the distance between any two distant objects can be measured.



REFLECTION OF A SOUND WAVE

Reflection off a Nearby Wall



Time (s): 0.00

Reflection off a Distant Cliff



ECHO

The sound heard after reflection from a rigid obstacle is called an echo.

REVERBERATION

The repeated reflection of sound waves that results in persistence of sound for sometimes.



REFLECTION OF A SOUND WAVE PRACTICAL APPLICATIONS



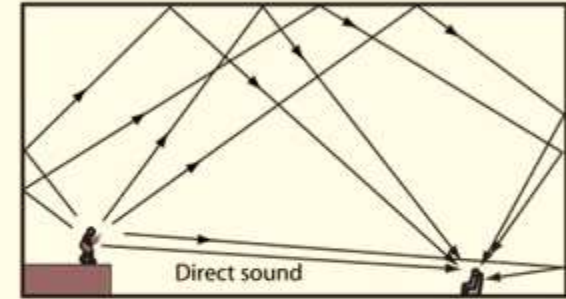
MEGAPHONE

Megaphone is a horn-shaped tube. The sound waves are prevented from spreading out by successive reflections and are confined to the air in the tube.



HEARING AIDS

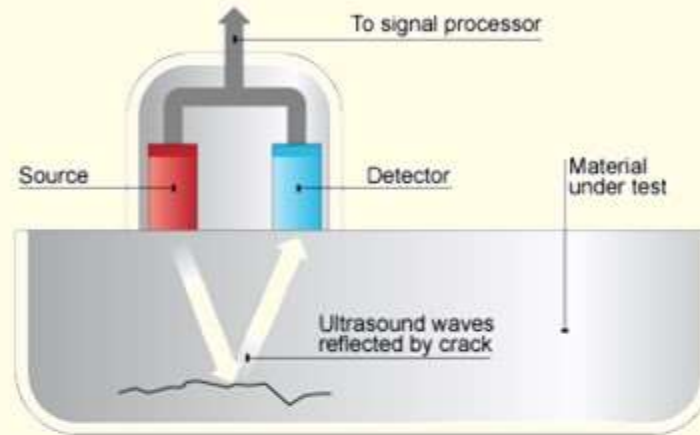
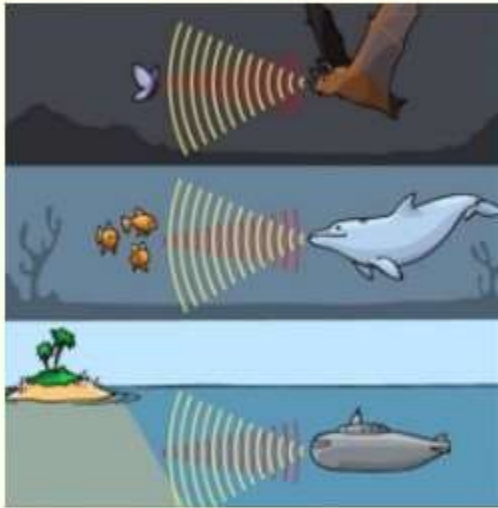
Here the sound waves, which are received by the hearing aid, are reflected into a narrower area leading to the ear.



SOUND BOARD

The reflection of sound waves is used in auditorium to spread the waves uniformly throughout the hall. Reflection of sound waves is done by using Sound Boards.

REFLECTION OF ULTRASOUND - APPLICATIONS



1. It is used for medical diagnosis and therapy and also as a surgical tool.
2. It is used in SONAR to measure the depth of the sea floor.
3. Ultrasound is also passed through concrete objects to detect any cracks inside the objects.
4. Bats and porpoises use ultrasound for navigation and to locate food in darkness.
5. It is used to detect defective foetus.
6. It is used as a tool in the treatment of muscular pain.
7. Sonogram (is a technique of 3-dimensional photographs with the help of ultrasonic waves) is used to locate the exact position of an eye tumor.

The sound waves that lies above our audible range is called ultrasound. It has a frequency of more than 20,000 Hz

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